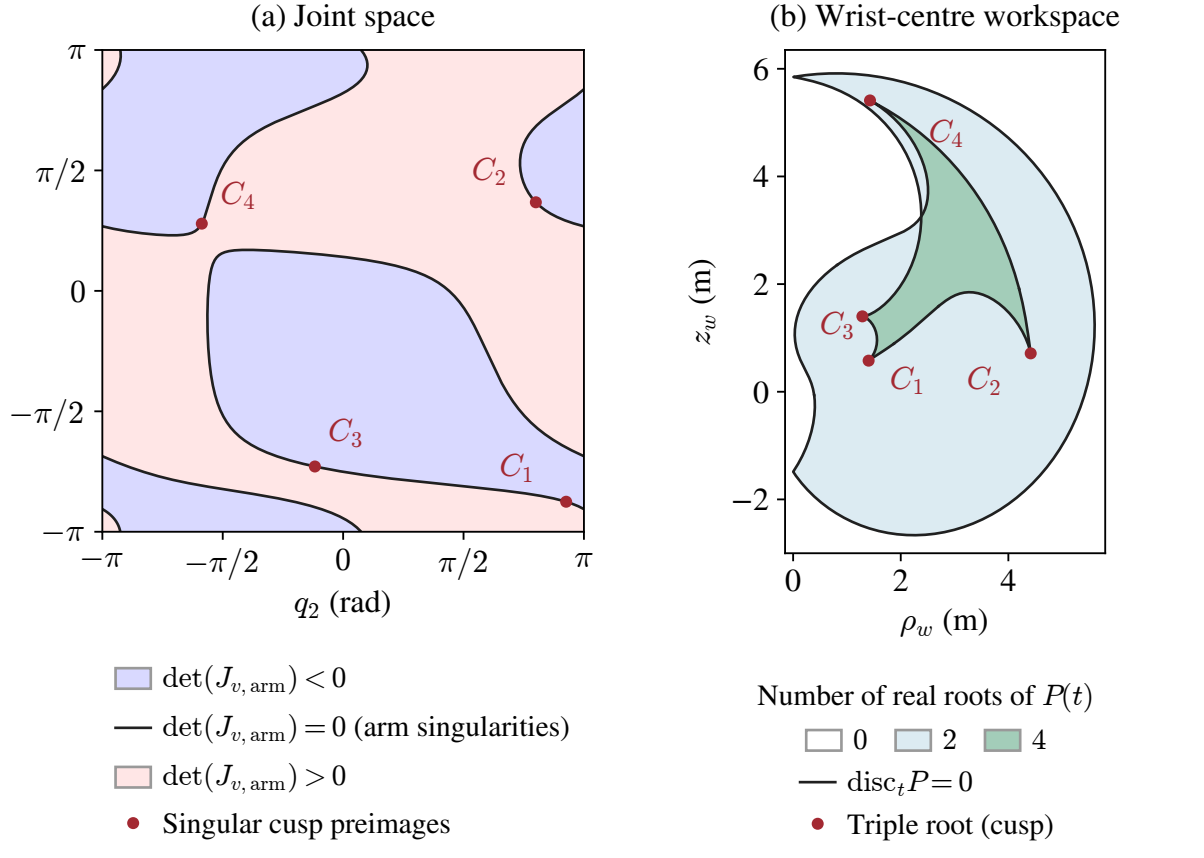

(a) Joint space
(b) Wrist-centre workspace
C_1
C_2
C_3
C_4
q_2 (rad)
ρ_w (m)
z_w (m)
$\det(J_{v,\mathrm{arm}}) < 0$
$\det(J_{v,\mathrm{arm}}) = 0$ (arm singularities)
$\det(J_{v,\mathrm{arm}}) > 0$
Singular cusp preimages
Number of real roots of $P(t)$
0
2
4
$\mathrm{disc}_t P = 0$
Triple root (cusp)